

Session Objectives

- Learn how to perform a neurological assessment of the Sensory System and the use of Special Techniques
- Learn how to obtain a detailed health history in a patient with a neurological complaint
- Review how to document the physical findings for a neurological examination and include the sensory system

Sensory Pathways

NEW YORK INSTITUTE
OF TECHNOLOGY

College of Osteopathic
Medicine

Sensory impulses:

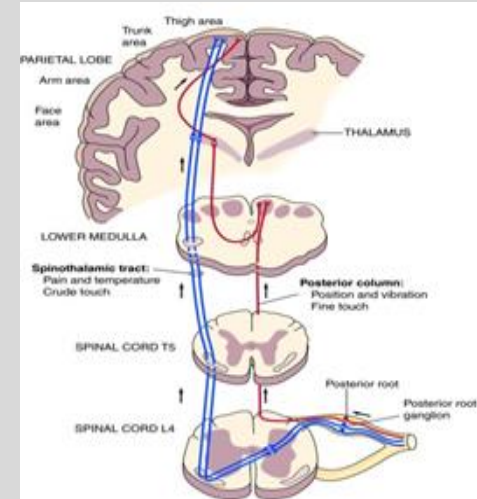
Participate in reflex activity, conscious sensation, locate body position in space, help regulate internal autonomic functions: heart rate, respiration and blood pressure

Reach the Sensory Cortex of the brain via

- Spinothalamic tract (smaller sensory neurons that are unmyelinated or thinly myelinated axons)- pain, temperature crude touch
- Posterior columns (larger neurons with heavily myelinated axons)- vibration, proprioception, kinesthesia, pressure, fine touch (skin and joint position receptors)

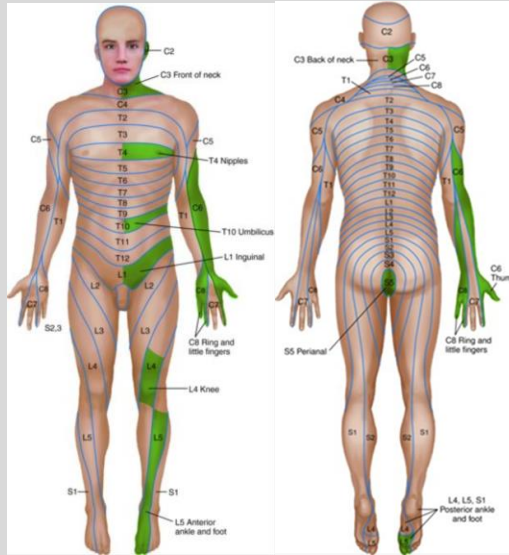
*Both the Spinothalamic tract and Posterior columns will have a cross-over to the opposite side as it travels to the Thalamus.

See diagram on slide- Spinothalamic tract (blue) and Posterior column (red)



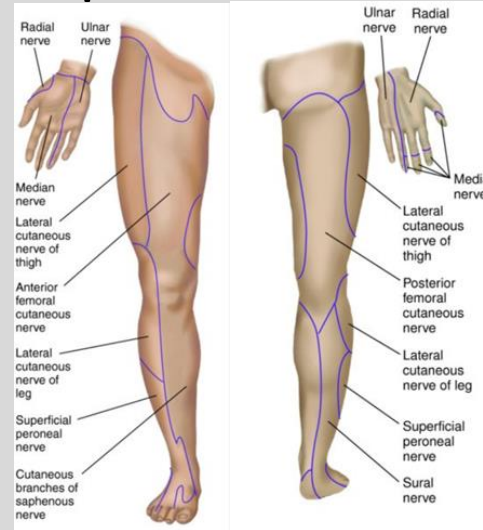
Sensory

Dermatomes



Dermatomes innervated by
Posterior roots

Peripheral Nerve Pattern



Areas innervated by
Peripheral nerves

Sensory Testing- Dermatomes

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

Pain

- Touch multiple areas of the body with a sharp/semi-sharp object or soft object and have the patient identify what it is and its location with their eyes closed



Sensory Testing

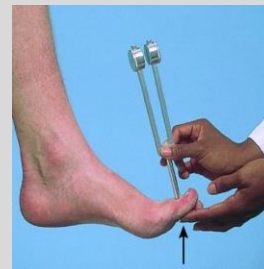
**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

Vibration

- Use a low pitched vibrating tuning fork (128 Hz)
- Place tuning fork over the distal interphalangeal joint (IP)- finger or toe
- If vibration is impaired test a more proximal bony prominence ex: wrist, elbow, medial malleolus, shin, patella, anterior superior iliac spine, spinous processes and clavicles

*Often the first sensation lost in peripheral neuropathy



Proprioception

Joint position sense

- Test fingers and toes first
- If distal joints are normal, most likely proprioception is intact in proximal joints



Discriminative Sensations

- Tests the ability of the sensory cortex to correlate, analyze and interpret sensations
- Touch and position sense need to be intact for tests
- Absence or a decrease in discriminative sensation indicates a lesion in the sensory cortex
- Patient's eyes should be closed for all tests

Discriminative Sensations

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

Stereognosis

- Identify object through touch



Graphesthesia (number identification)

- Identify the number drawn on patient's palm



Two Point Discrimination

- Touch two places simultaneously using open ends of paper clip
- Normal minimal distance for discrimination varies for different parts of the body



Discriminative Sensations

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

Point Localization

- Patient has eyes closed
- Briefly touch a point on the patient's skin
- Ask the patient to open their eyes and point to the place touched
- If the patient has a lesion of the sensory cortex, their ability to localize the point(s) is impaired



Extinction

- Patient has eyes closed
- Simultaneously stimulate areas on the both sides of the body.
- Patient will identify the location of the stimuli

With Sensory Cortex lesions, only one stimulus may be felt. The stimulus that is not recognized, is “extinguished”, and is on the contralateral side of the affected sensory cortex.



Brudzinski and Kernig Sign

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

Brudzinski Sign

- Patient's head and neck are flexed forward
- Normal response- Hips and knees should remain relaxed
- Positive Brudzinski Sign- Flexion of both hips and knees



Kernig Sign

- Patient's leg is flexed at the hip and knee
- Slowly extend the leg and straighten the knee
- Positive Kernig's Sign- Pain and increased resistance to knee extension



Axial Compression Test

Test for cervical neck impingement

- Patient is seated
- Examiner places both hands on top of the patient's head
- Exert a downward force with both hands

*Test is positive if pain/neurological symptom are reproduced in the cervical spine and/or upper extremity

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine



Foraminal Distraction Test

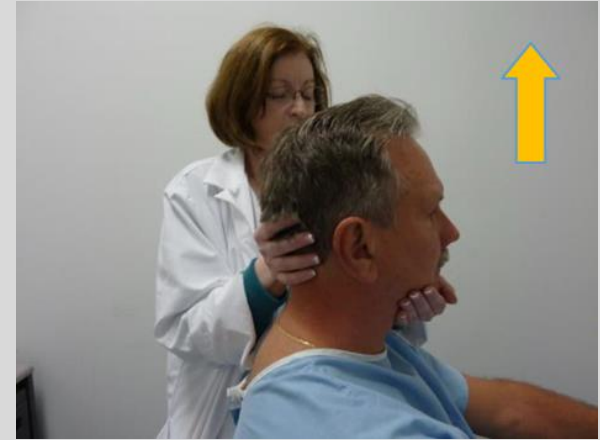
Test for cervical nerve impingement

- Patient is seated
- Examiner places one hand under the patient's chin and the other hand under the occiput
- An upward traction force is applied on the cervical spine

*Test is positive if the pain/neurological symptoms of the cervical spine and/or upper extremity are reduced with the traction force

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine



Spurling Test

Test for Cervical Nerve Root Compression

- Patient looks over their shoulder and then up at the ceiling
- Downward pressure is applied to the patient's head
- Maneuver reproduces the neck pain with radiation to the arm of the same side that the head is turned to
If there is pain with just the positioning of the head
It is not necessary to apply downward pressure of the head

NEW YORK INSTITUTE
OF TECHNOLOGY

College of Osteopathic
Medicine



Straight Leg Raise (with Bragard's Test)

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

Tests for radicular sciatic nerve pain

- Patient is supine with legs extended
- Test unaffected side first
- Raise the patient's relaxed and straightened leg by the ankle and place a hand on the knee to prevent it from bending.
- Stop when the patient feels pain or maximal flexion has been achieved. (60-120 degrees)
- Repeat on affected side.

Bragard's Test- lower the leg to the point just below the level of pain and add dorsiflexion of the foot
L5 portion of the sciatic nerve will be stretched and cause pain that radiates below the knee



Well (Opposite) Straight Leg Raising (Crossed Straight Leg Raise)

Test differentiates between sciatic nerve irritation and a herniated disc space occupying lesion

- Patient is supine with legs extended
- Examiner asks the patient to raise their fully extended leg on the uninvolved
- * The test may indicate a possible space occupying lesion if the pain occurs on the opposite side of the leg that was raised



Hoover Test

Test for malingering

- Patient is supine
- Unilateral exam
- Examiner places their hands under the patient's calcaneus bilaterally (see photo)
- Patient is instructed to raise the leg of the involved side without bending their knee
- Examiner is monitoring for a downward pressure of the heel on the side not being tested

*Test is positive if there is an absence of a downward pressure of the uninvolved foot when the patient is raising the involved leg

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine



Neurological Exam

Start with the patient's history

Common or Concerning Symptoms:

- Headache
- Dizziness/Vertigo
- Weakness
- Numbness, Abnormal or Absent Sensation
- Syncope/ Near Syncope
- Seizures
- Tremors/ Involuntary Movements

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

Common Neurological Complaints

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

- **Headache**
 - OPQRSTA
 - Onset, History of head trauma
 - Severity *very important to assess “worst headache of my life”
 - Location
 - Duration
 - Aura or Prodrome
 - Associated symptoms
 - Visual changes-double vision
 - Photo or Phonophobia
 - Weakness
 - Loss of sensation
 - Worse with coughing, sneezing, head movement (alteration of intracranial pressure)
 - Infectious symptoms: fever, stiff neck, sinus congestion, sore throat, ear pain
 - Take a thorough past history also including medications (headache could be a side effect or a withdrawal symptom)
 - Lifestyle changes: increased/decreased caffeine, nicotine, alcohol, stress
 - Recommend patient keep a headache diary to note triggers

Common Neurological Complaints

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

- **Near-syncope and Syncope**
 - OPQRSTA
 - Important to assess “Was there loss of consciousness?”
 - Near-Syncope
 - Hear external noises or voices throughout episode
 - Lightheaded
 - Weakness
 - Syncope
 - Complete loss of consciousness
 - What were you doing at time of onset?
 - Standing, sitting or lying down?
 - Any triggers or warning symptoms?
 - How long did the episode last?
 - How quickly did the onset occur?
 - Was the episode witnessed?
 - Associated symptoms
 - Palpitations- cardiac arrhythmia
 - Diaphoretic
 - Nausea
 - Pallor
 - Loss of bowel/bladder control
 - Take a thorough past history including PMH: heart disease, seizures and medications-

Bates, 13th Edition Chapter 24

Tics and Involuntary Movements

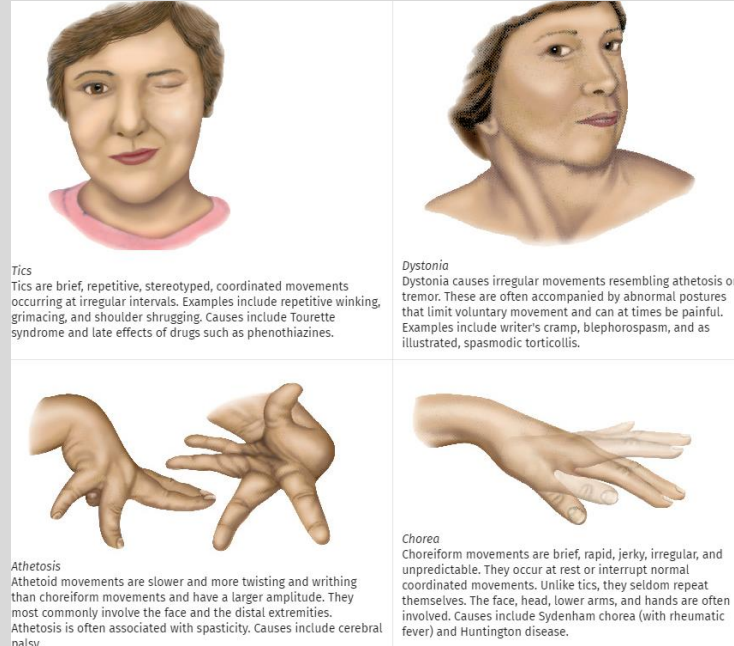
NEW YORK INSTITUTE
OF TECHNOLOGY

College of Osteopathic
Medicine



Oral-Facial Dyskinesias

Oral-facial dyskinesias are arrhythmic, repetitive, bizarre movements that chiefly involve the face, mouth, jaw, and tongue: grimacing, pursing of the lips, protrusions of the tongue, opening and closing of the mouth, and deviations of the jaw. The limbs and trunk are involved less often. These movements may be a late complication of antipsychotic or antiemetic drugs such as phenothiazines, termed *tardive* (late) dyskinesias. They also occur in long-standing psychoses, in some older adults, and in some edentulous persons.



Screening for Diabetic Peripheral Neuropathy

NEW YORK INSTITUTE
OF TECHNOLOGY

College of Osteopathic
Medicine

- Diabetic patients are unaware of neuropathy
- Unaware of sensory loss
- Distal symmetric polyneuropathy is the most common neuropathy for these patients
 - Slowly progressive and often asymptomatic
 - Risk factor for ulceration, arthropathy and amputation
 - Prevalence- 20% in long term type I diabetic patients
 - 50% in long term type II diabetic patients
- Important to maintain tight glycemic control, routine foot examinations assessing for neuropathy (including- Temperature or Pinprick sensation, Proprioception, ankle reflexes, vibration perception (128 Hz tuning fork) and plantar light touch sensation using a 10 –G monofilament), skin breakdown, poor circulation and musculoskeletal abnormalities

Documentation of a Normal Neurological Examination

Neuro-

Mental Status: The patient is alert, attentive, and oriented to person, place and time. Speech; clear and fluent with normal comprehension

- CN I not tested CN II-XII grossly intact
- Motor strength is 5/5 in all extremities, good muscle bulk and tone
- Rapid alternating movements and finger to nose are intact
- Gait is normal, no evidence of ataxia
- Negative Romberg and Pronator Drift
- Sensory- Pinprick, light touch, position and vibration intact
- Deep Tendon Reflexes 2+ bilaterally, plantar reflexes downgoing or plantar flexion bilaterally

Summary

- Importance of taking a good history when evaluating a neurological problem
- A neurological diagnosis can be obtained through performing a thorough examination that includes testing the motor and sensory systems, coordination, gait and reflexes in addition to the cranial nerves.
- Importance to know how to document the neurological examination.